

COMP 3311

Database Management Systems

Lab 2

Oracle Database, SQL*Plus and
Oracle SQL Developer

Lab Topics

- ❑ What is **Oracle Database**, **SQL*Plus**, and an **SQL*Plus** script file.
- ❑ How to connect to **Oracle Database** using **Oracle SQL Developer**.
- ❑ How to create and execute **SQL*Plus** script files in **Oracle SQL Developer**.
- ❑ How to create, modify and list the contents of an **Oracle Database** table.
- ❑ How to change your Oracle account password.

Why Oracle Database?

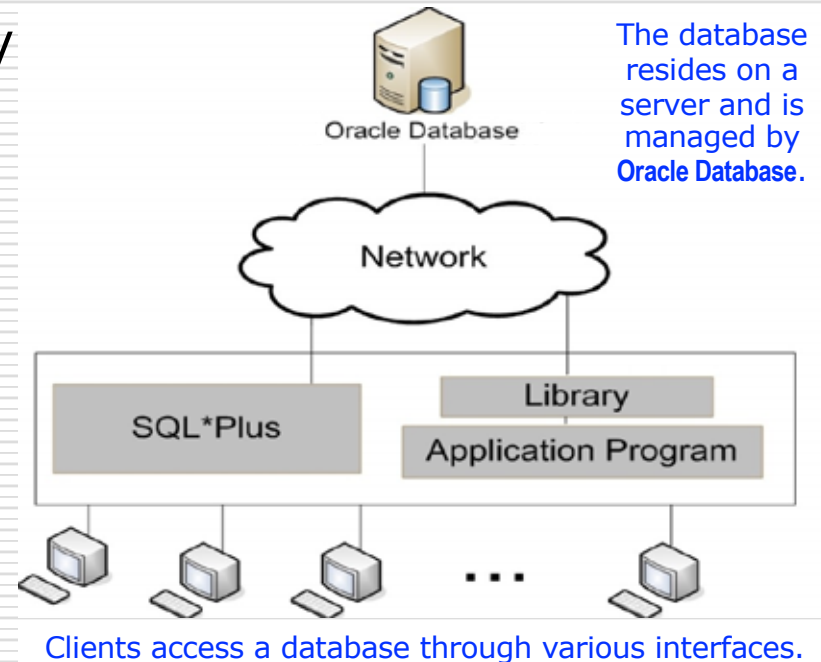
- ❑ **Oracle Database** is one of the most widely used commercial DBMSs; you are likely to use it at some point in the future.
- ❑ Other relational DBMSs are very similar to **Oracle Database**.
- ❑ You should be able to use any other relational DBMS if you are familiar with **Oracle Database** as they all use the **SQL** language.

Oracle Database

- ❑ The first commercially available relational DBMS.
- ❑ The CSE labs provide **Oracle Database 19c**.
Earlier versions can also be used for the course.
- ❑ If you are so inclined, you can download the free **Oracle Database Express Edition (XE)** to install on your own computer from
<https://www.oracle.com/database/technologies/xe-downloads.html>
 - Requires registration/login; only Windows, Linux available.
(Sorry Mac users; you should complain to Oracle.)
 - Instructions for installing **Oracle Database Express Edition (XE)** on Windows can be found at the end of these lab notes.

The Oracle Client/Server Model

- ❑ Clients accept SQL statements/commands from users and send them to the **Oracle Database** server over a network.
- ❑ The **Oracle Database** server executes the queries and returns the results to the clients, which then deliver the results to the user.

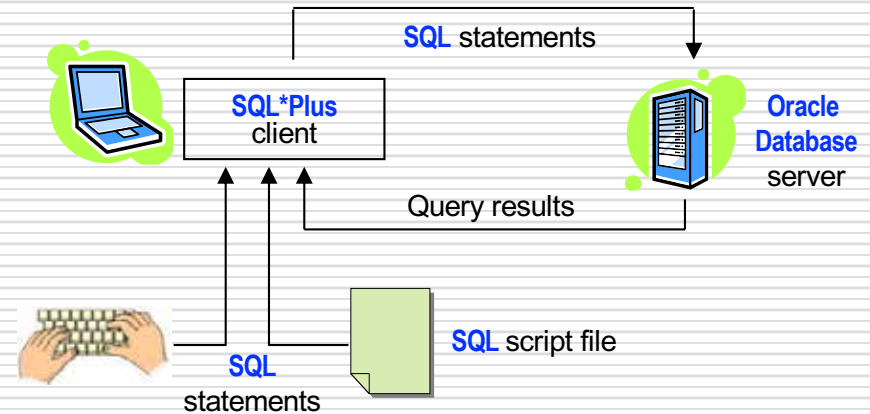


- ❑ The **Oracle Database** server for this course is **dbsvr1.cse.ust.hk**.

The **Oracle Database** server can be accessed directly only from the CSE Department network and the campus network (with IP addresses starting with 143.89). From other networks, including all Wi-Fi networks, it needs to be accessed through the HKUST VPN (see <http://itsc.ust.hk/apps/vpn/> for how to connect to the HKUST VPN).

SQL*Plus

- ❑ **SQL*Plus** is an interactive and batch query tool that enables **SQL**, **PL/SQL**, **SQL*Plus** and operating system commands to be executed.



- ❑ **SQL*Plus** allows users to:
 - format, perform calculations on, store, and print query results;
 - examine table definitions;
 - develop and run SQL script files;
 - perform database administration.
- ❑ The **SQL*Plus** client can be run from within **Oracle SQL Developer** or directly from an operating system command line.

Oracle SQL Developer

- ❑ **Oracle SQL Developer** provides a desktop-like interface to the **SQL*Plus** client allowing users to:
 - browse, create, edit, and delete tables (i.e., relations);
 - run **SQL** statements and script files;
 - edit and debug **PL/SQL** code;
 - manipulate and export data;
 - view and create reports.

Reminder: Download **Oracle SQL Developer** from

<https://www.oracle.com/tools/downloads/sqldev-downloads.html>

Requires registration/login; Windows, Mac, Linux available (requires JDK 11).

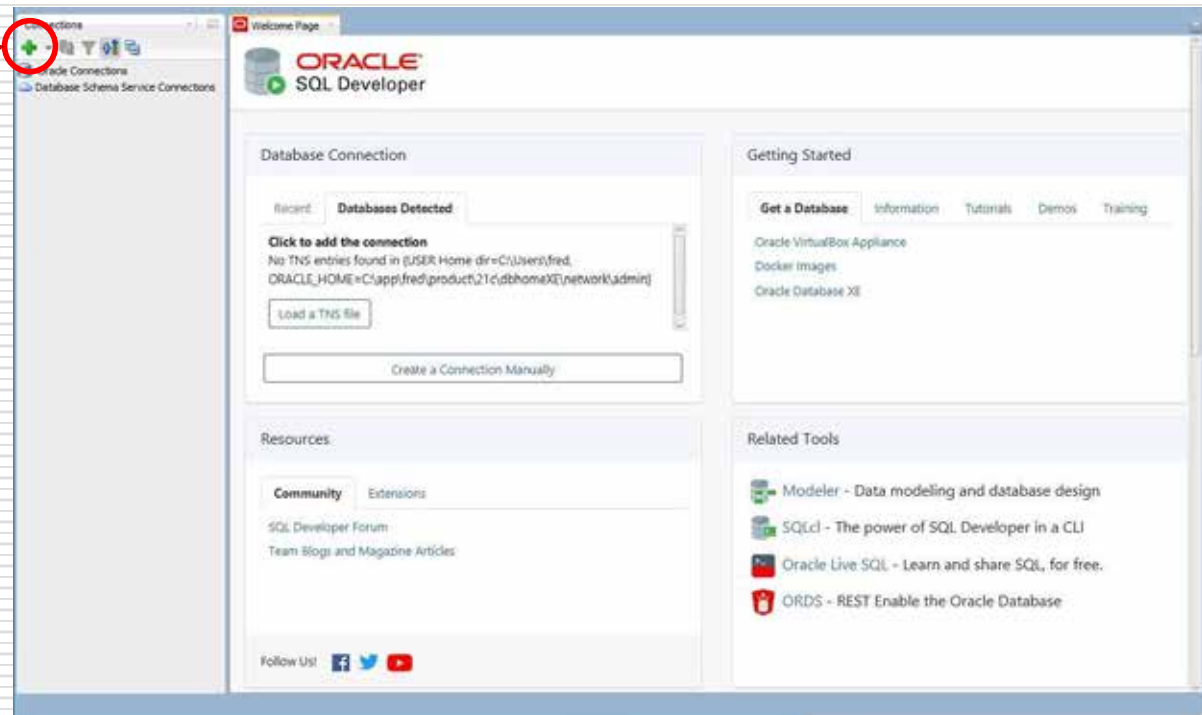
Connecting To Oracle Database Using Oracle SQL Developer (1)

1. Run the program "sqldeveloper"¹.
2. Select the green "+" symbol in the **Connections** navigator (left-side) pane. (If the **Connections** navigator is not visible, select **View**→**Connections** in the **Oracle SQL Developer** menu bar.)

The **New / Select Database Connection** dialog appears as shown on the next slide.

1. In Lab 4 (room 4210), you can search for "sql" and double click "sqldeveloper" in the search results.

New
Connection



Connecting To Oracle Database Using Oracle SQL Developer (2)

3. Enter/select the information outlined below in red using your Oracle username and Oracle password for the Username and Password fields, respectively.

(Note: these are not the same as your UST/CSE username and password.)

The screenshot shows the 'New / Select Database Connection' dialog box. The 'Name' field is highlighted with a red box and contains the text 'a meaningful connection name'. The 'Database Type' is set to 'Oracle'. The 'User Info' tab is selected, showing 'Proxy User'. The 'Authentication Type' is set to 'Default'. The 'Username' field is highlighted with a red box and contains the text 'your Oracle username'. The 'Password' field is highlighted with a red box and contains the text 'your Oracle password'. The 'Role' is set to 'default'. The 'Save Password' checkbox is checked. The 'Connection Type' is set to 'Basic'. The 'Details' tab is selected, showing 'Advanced'. The 'Hostname' field is highlighted with a red box and contains the text 'dbsvr1.cse.ust.hk'. The 'Port' field is highlighted with a red box and contains the text '1521'. The 'Service name' radio button is selected, and the 'Service name' field is highlighted with a red box and contains the text 'comp3311.cse.ust.hk'. The 'Status' field is empty. The 'Help', 'Save', 'Clear', 'Test', 'Connect', and 'Cancel' buttons are at the bottom.

Your Oracle username and password were sent to you by email. An Oracle username has the form "comp3311stuXXX" where XXX are digits.

Connecting To Oracle Database Using Oracle SQL Developer (3)

4. Select the **Test** button (at the bottom middle of the **New / Select Database Connection** dialog) to check that the information entered is correct.

You should see the message **Success** in the **Status** field near the lower left corner of the **New / Select Database Connection** dialog just above the **Help** button.

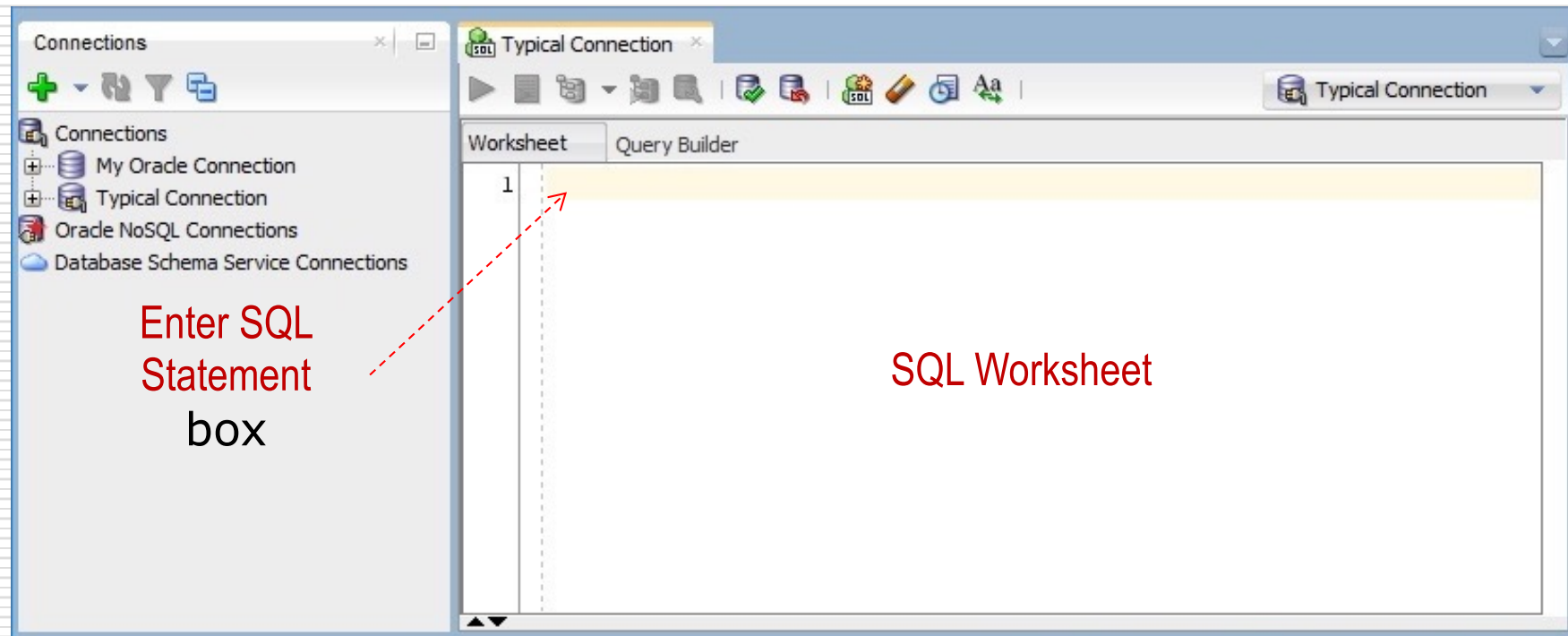
Correct any errors and test the connection again.

5. Select the **Save** button to save the connection information for future use.

6. Select the **Connect** button.

You should see an **SQL Worksheet** like that shown on the next slide.

Connecting To Oracle Database Using Oracle SQL Developer (4)



SQL Worksheet

- ❑ An **SQL Worksheet** is used to enter and execute **SQL**, **PL/SQL** and **SQL*Plus** statements.
- ❑ There can be multiple worksheets open for a given connection.
- ❑ To create an **SQL Worksheet**:
 - right-click a connection in the **Connections** navigator and select **Open SQL Worksheet**, or
 - select **Tools** and then **SQL Worksheet**, or
 - select  (the **SQL Worksheet** button) in the toolbar. This button also allows you to select the connection to use for the worksheet.

SQL Worksheet Toolbar



- The **SQL Worksheet** toolbar contains the following buttons (among others).

-  **Run Statement** executes a *single statement at the cursor* or *several selected statements* in the **Enter SQL Statement** box.
-  **Run Script** executes *all statements* in the **Enter SQL Statement** box using the **Script Runner**.
-  **Commit** writes any changes to the database, ends the transaction and clears the **Query Result** and **Script Output** tabs.
-  **Rollback** discards any changes without writing them to the database, ends the transaction and clears the **Query Result** and **Script Output** tabs.
-  **Clear** erases the statements in the **Enter SQL Statement** box.

Opening And Executing A Script File

□ To open a script file:

1. select File→Open in the **Oracle SQL Developer** menu bar or select  (the **Open** button) in the toolbar;
2. in the **Open** dialog, navigate to the script file;
3. double-click the script file or select it and select the **Open** button.

Alternatively, you can drag and drop a script file onto a worksheet in **Oracle SQL Developer**¹.

□ To run (execute) a script file, select  (the **Run Script** button) shown in the example script on the next slide.

1. This is only possible if there is already a worksheet open.

Example Script File

Run Script

Note: When you run the `lab2db.sql` script *the first time* you will get the error message:

Error starting at line : 9 in command -
drop table Student

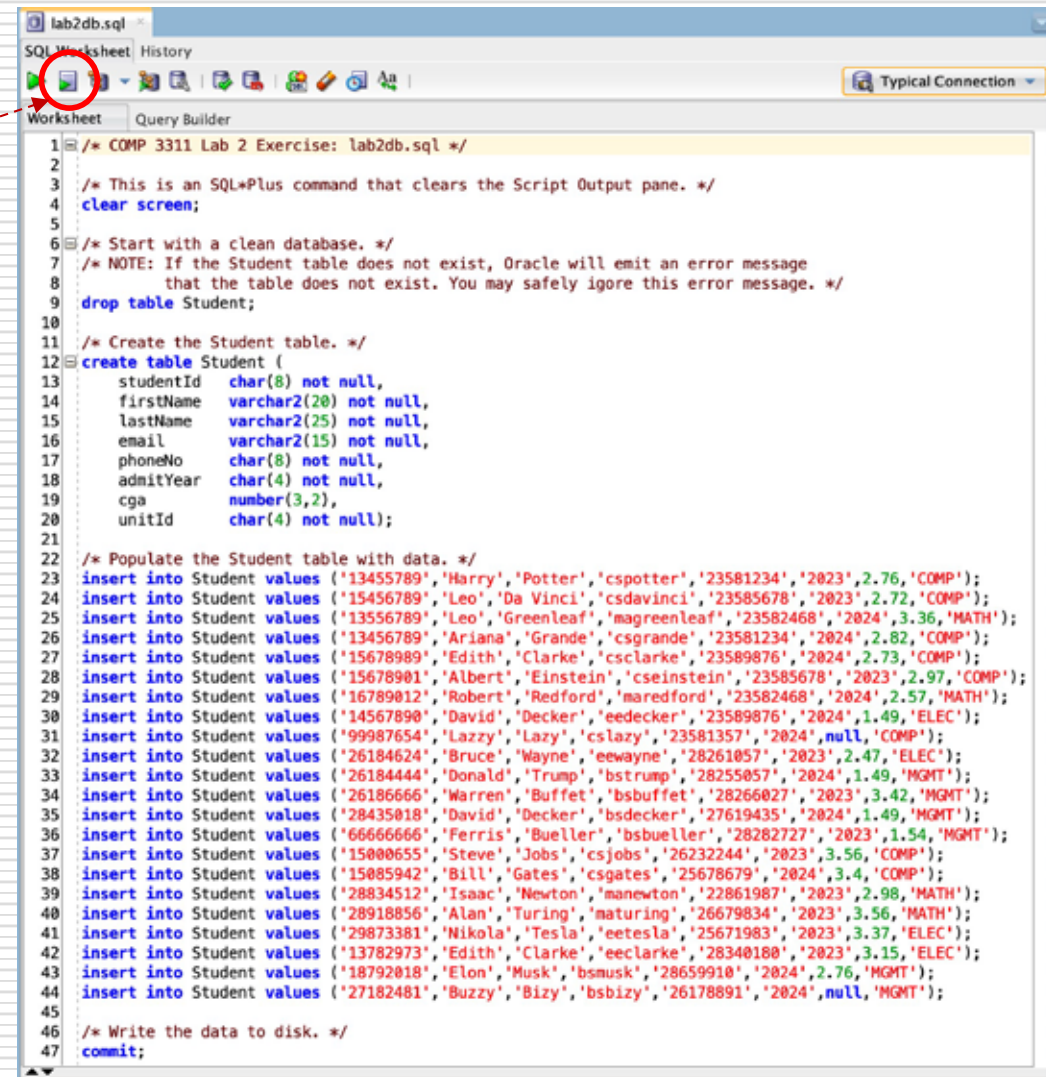
Error report -

ORA-00942: table or view does not exist
00942. 00000 - "table or view does not exist"

*Cause:

*Action:

You may ignore this message. It arises because the script first drops (deletes) the Student table, but no such table exists before the script is executed. If you run the script again, you will not get this error as the Student table now exists.



```
1  /* COMP 3311 Lab 2 Exercise: lab2db.sql */
2
3  /* This is an SQL*Plus command that clears the Script Output pane. */
4  clear screen;
5
6  /* Start with a clean database. */
7  /* NOTE: If the Student table does not exist, Oracle will emit an error message
8   that the table does not exist. You may safely ignore this error message. */
9  drop table Student;
10
11 /* Create the Student table. */
12 create table Student (
13     studentId    char(8) not null,
14     firstName    varchar2(20) not null,
15     lastName     varchar2(25) not null,
16     email        varchar2(15) not null,
17     phoneNo      char(8) not null,
18     admitYear    char(4) not null,
19     cga          number(3,2),
20     unitId       char(4) not null;
21
22 /* Populate the Student table with data. */
23 insert into Student values ('13455789','Harry','Potter','cspotter','23581234','2023',2.76,'COMP');
24 insert into Student values ('15456789','Leo','Da Vinci','csdavinci','23585678','2023',2.72,'COMP');
25 insert into Student values ('13556789','Leo','Greenleaf','magreenleaf','23582468','2024',3.36,'MATH');
26 insert into Student values ('13456789','Ariana','Grande','csgrande','23581234','2024',2.82,'COMP');
27 insert into Student values ('15678989','Edith','Clarke','csclarke','23589876','2024',2.73,'COMP');
28 insert into Student values ('15678901','Albert','Einstein','cseinstein','23585678','2023',2.97,'COMP');
29 insert into Student values ('16789012','Robert','Redford','maredford','23582468','2024',2.57,'MATH');
30 insert into Student values ('14567890','David','Decker','eedecker','23589876','2024',1.49,'ELEC');
31 insert into Student values ('99987654','Lazy','Lazy','cslazy','23581357','2024',null,'COMP');
32 insert into Student values ('26184624','Bruce','Wayne','eewayne','28261057','2023',2.47,'ELEC');
33 insert into Student values ('26184444','Donald','Trump','bs Trump','28255057','2024',1.49,'MGMT');
34 insert into Student values ('26186666','Warren','Buffet','bsbuffet','28266027','2023',3.42,'MGMT');
35 insert into Student values ('28435018','David','Decker','bsdecker','27619435','2024',1.49,'MGMT');
36 insert into Student values ('66666666','Ferris','Bueller','bsbueller','28282727','2023',1.54,'MGMT');
37 insert into Student values ('15000655','Steve','Jobs','csjobs','26232244','2023',3.56,'COMP');
38 insert into Student values ('15085942','Bill','Gates','csgates','25678679','2024',3.4,'COMP');
39 insert into Student values ('28834512','Isaac','Newton','manewton','22861987','2023',2.98,'MATH');
40 insert into Student values ('28918856','Alan','Turing','maturing','26679834','2023',3.56,'MATH');
41 insert into Student values ('29873381','Nikola','Tesla','eetesla','25671983','2023',3.37,'ELEC');
42 insert into Student values ('13782973','Edith','Clarke','eeclarke','28340180','2023',3.15,'ELEC');
43 insert into Student values ('18792018','Elon','Musk','bsmusk','28659910','2024',2.76,'MGMT');
44 insert into Student values ('27182481','Buzzy','Bizy','bsbizy','26178891','2024',null,'MGMT');
45
46 /* Write the data to disk. */
47 commit;
```

SQL DDL Commands in the Script File

SQL command: `drop table` *<tablename>*;

- Deletes a table (both `data` *and* `schema`).

SQL command: `create table` *<tablename>*
(
 column1 datatype [`null` | `not null`],
 column2 datatype [`null` | `not null`], ...
);

- Creates a new table in the database and defines the columns, data type and optional null constraint for each column.

SQL command: `insert into` *<tablename>* `values` (*column1value, column2value, ...*);

- Adds one or more records to a table with a value for each column of *<tablename>*.

Script Output Tab

- ❑ The **Script Output** tab displays the **result of executing a script file**; its toolbar has the following buttons:



Pin keeps the tab's contents in the window when another object is selected in the **Connections** navigator.



Clear erases the tab's contents.

Note: It is strongly recommended to clear a tab's contents before running a script again to avoid becoming confused by seeing the results of previous executions.



Save File saves the tab's contents to a file.



Print sends the tab's contents to a printer.

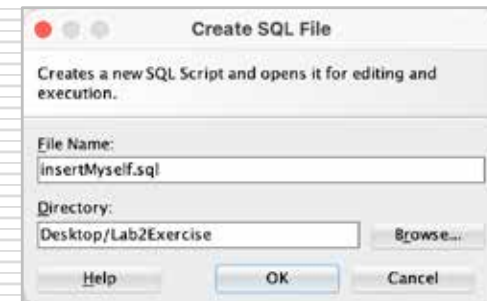
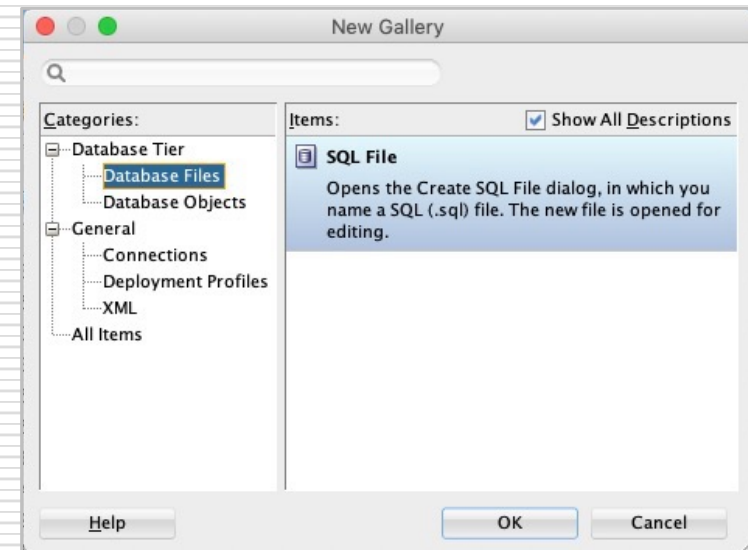


Run Script Output as Script executes the tab's contents as a script file.



Creating A Script File

1. Select File→New in the **Oracle SQL Developer** menu bar.
2. In the **New Gallery** dialog, select **Database Files** in the **Categories** pane; select the **OK** Button.
3. In the **Create SQL File** dialog, enter a name in the **File Name** text box.
4. Navigate to or enter the name of the **Directory** where the script file should be stored.
5. Select the **OK** button.

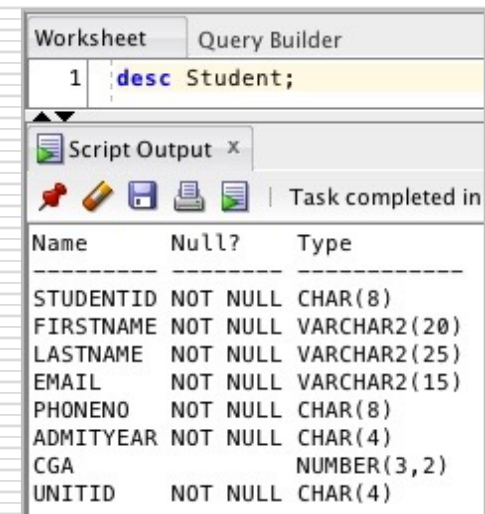


Displaying The Structure Of A Table (1)

SQL command: `desc[ribe] <tablename>;`

1. Open a new **SQL Worksheet** and connect to **Oracle Database** using your Oracle connection.
2. If you have not yet done so, execute the **lab2db.sql** script file.
3. In the **Enter SQL Statement** box type
 `desc Student;`
as shown in the figure.
4. Select  (the **Run Statement** button).

The **Script Output** tab displays the result as shown in the figure.



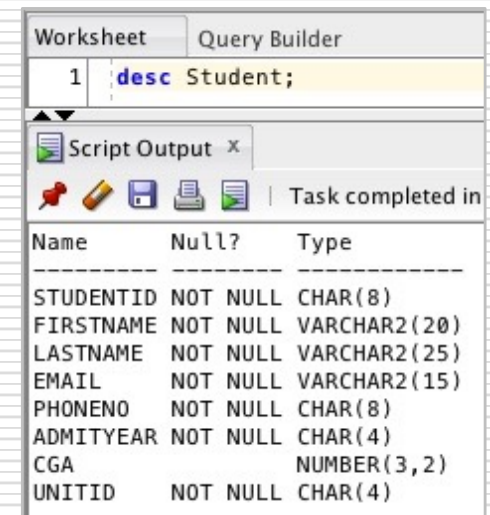
The screenshot shows a software window with two tabs: 'Worksheet' and 'Query Builder'. The 'Worksheet' tab is active, displaying the SQL command 'desc Student;' in a text area. Below the text area is a 'Script Output' tab, which is also active and displays the result of the command. The result is a table with three columns: 'Name', 'Null?', and 'Type'. The table lists the columns of the 'Student' table: STUDENTID, FIRSTNAME, LASTNAME, EMAIL, PHONENO, ADMITYEAR, CGA, and UNITID, along with their nullability and data types.

Name	Null?	Type
STUDENTID	NOT NULL	CHAR(8)
FIRSTNAME	NOT NULL	VARCHAR2(20)
LASTNAME	NOT NULL	VARCHAR2(25)
EMAIL	NOT NULL	VARCHAR2(15)
PHONENO	NOT NULL	CHAR(8)
ADMITYEAR	NOT NULL	CHAR(4)
CGA		NUMBER(3,2)
UNITID	NOT NULL	CHAR(4)

Displaying The Structure Of A Table (2)

❑ The **Script Output** tab shows:

- **Name** – The name of the attribute.
- **Null?** – Indicates whether a column must contain data.
- **Type** – The data type of the column.
 - ❑ **CHAR(s)** – A character string of fixed length **s**.
 - ❑ **VARCHAR2(s)** – A variable length character string of maximum length **s**.
 - ❑ **NUMBER(p, s)** – A number with a total of **p** digits with **s** digits to the right of the decimal point.

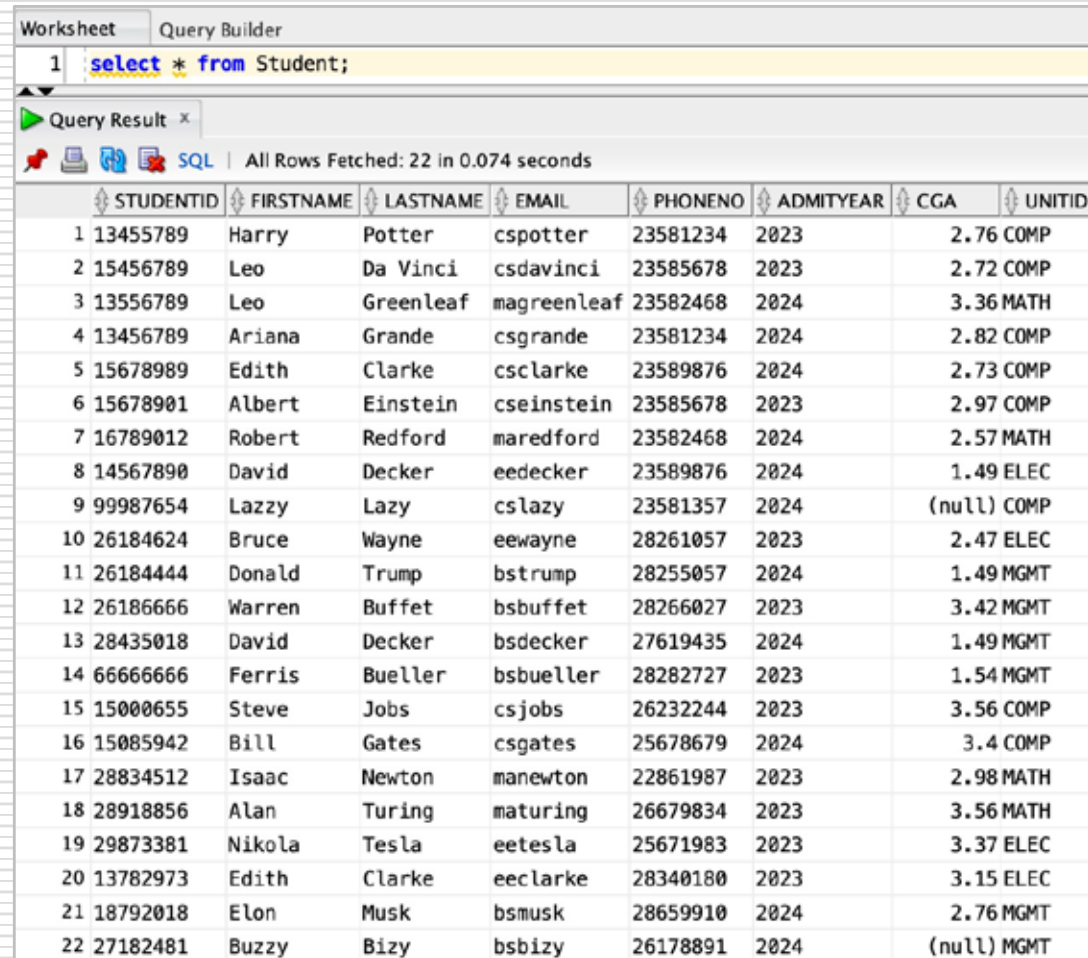


Name	Null?	Type
STUDENTID	NOT NULL	CHAR(8)
FIRSTNAME	NOT NULL	VARCHAR2(20)
LASTNAME	NOT NULL	VARCHAR2(25)
EMAIL	NOT NULL	VARCHAR2(15)
PHONENO	NOT NULL	CHAR(8)
ADMITYEAR	NOT NULL	CHAR(4)
CGA		NUMBER(3,2)
UNITID	NOT NULL	CHAR(4)

Displaying The Contents Of A Table

SQL command: `select * from <tablename>;`

- In the **Enter SQL Statement** box, type
`select * from Student;`
as shown in the figure.
- **Select**  (the **Run Statement** button).
- The **Query Result** tab displays the **SQL** statement result as shown in the figure.



The screenshot shows a database query tool interface. The 'Query Builder' tab is active, displaying the SQL command `select * from Student;` in a text box. Below the text box, the 'Query Result' tab is selected, showing a table with 22 rows of data. The table has columns for STUDENTID, FIRSTNAME, LASTNAME, EMAIL, PHONENO, ADMITYEAR, CGA, and UNITID. The data is sorted by STUDENTID in ascending order.

	STUDENTID	FIRSTNAME	LASTNAME	EMAIL	PHONENO	ADMITYEAR	CGA	UNITID
1	13455789	Harry	Potter	cspotter	23581234	2023	2.76	COMP
2	15456789	Leo	Da Vinci	csdavinci	23585678	2023	2.72	COMP
3	13556789	Leo	Greenleaf	magreenleaf	23582468	2024	3.36	MATH
4	13456789	Ariana	Grande	csgrande	23581234	2024	2.82	COMP
5	15678989	Edith	Clarke	csclarke	23589876	2024	2.73	COMP
6	15678901	Albert	Einstein	cseinstein	23585678	2023	2.97	COMP
7	16789012	Robert	Redford	maredford	23582468	2024	2.57	MATH
8	14567890	David	Decker	eedecker	23589876	2024	1.49	ELEC
9	99987654	Lazzy	Lazy	cslazy	23581357	2024	(null)	COMP
10	26184624	Bruce	Wayne	eewayne	28261057	2023	2.47	ELEC
11	26184444	Donald	Trump	bstrump	28255057	2024	1.49	MGMT
12	26186666	Warren	Buffet	bsbuffet	28266027	2023	3.42	MGMT
13	28435018	David	Decker	bsdecker	27619435	2024	1.49	MGMT
14	66666666	Ferris	Bueller	bsbueller	28282727	2023	1.54	MGMT
15	15000655	Steve	Jobs	csjobs	26232244	2023	3.56	COMP
16	15085942	Bill	Gates	csgates	25678679	2024	3.4	COMP
17	28834512	Isaac	Newton	manewton	22861987	2023	2.98	MATH
18	28918856	Alan	Turing	maturing	26679834	2023	3.56	MATH
19	29873381	Nikola	Tesla	eetesla	25671983	2023	3.37	ELEC
20	13782973	Edith	Clarke	eeclarke	28340180	2023	3.15	ELEC
21	18792018	Elon	Musk	bsmusk	28659910	2024	2.76	MGMT
22	27182481	Buzzy	Bizy	bsbizy	26178891	2024	(null)	MGMT

Query Result Tab

- ❑ The **Query Result** tab displays the **result of executing a single SQL statement**; its toolbar contains the following buttons:
 -  **Pin** keeps the tab's contents in the window when another object is selected in the **Connections** navigator.
 -  **Print** sends the tab's contents to a printer.
 -  **Refresh** executes the SQL statement again to refresh the result.
 -  **Delete Persisted Settings** delete persisted settings such as any sort order specifications.
 -  **SQL** displays the SQL statement that produced the result in a new window.

Change Oracle Database Password

- To change your **Oracle Database** password, type

```
alter user <username> identified by "<new_password>";
```

in the **Enter SQL Statement** box and select  (the **Run Statement** button) where **<username>** and **<new_password>** are replaced with your Oracle username and your new password. Remember to add a **“;”** at the end of the SQL statement, because all SQL statements end with a **“;”**.

The following example changes the password to **123456** (**not** a good choice!):

```
alter user comp3311stu000 identified by "123456";
```

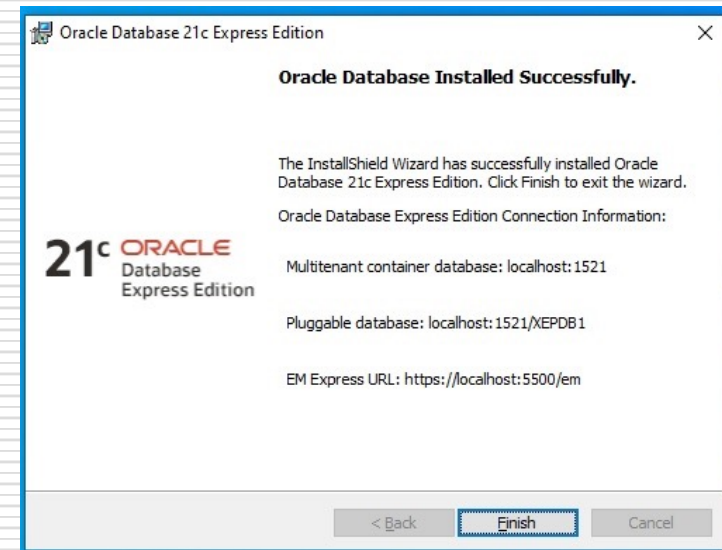
NOTE: DO NOT USE SPECIAL CHARACTERS IN YOUR PASSWORD!

Please remember your new password!

Installing Oracle Database Express Edition (XE)

Follow these instructions only if you wish to install Oracle XE.

1. Download **Oracle Database Express Edition (XE)** from <https://www.oracle.com/database/technologies/xe-downloads.html>
2. Run **setup** inside the downloaded folder to install **Oracle XE**.
(**Note**: Run **setup** again if the installation fails.)
3. When installation completes, note the information shown in the **InstallShield Wizard** dialog as shown on the right for future reference.
4. In **Oracle SQL Developer**, create a connection to your **Oracle XE** database as the DBA as shown in the example on the next slide.



Connect To Oracle Database Express Edition (XE) As DBA Using Oracle SQL Developer

Enter or select the information outlined below in red.
Leave all other information as shown.

The screenshot shows the Oracle SQL Developer connection configuration dialog. The 'Name' field is highlighted with a red box and contains the text 'a meaningful connection name'. The 'Database Type' is set to 'Oracle'. The 'User Info' tab is selected, showing the 'sys' user and 'SYSDBA' role. The 'Username' field is highlighted with a red box and contains 'sys'. The 'Password' field is highlighted with a red box and contains 'the DBA password created during installation'. The 'Role' dropdown is highlighted with a red box and shows 'SYSDBA'. The 'Authentication Type' is set to 'Default'. The 'Save Password' checkbox is checked. The 'Connection Type' is set to 'Basic'. The 'Details' tab is selected, showing the 'localhost' hostname, '1521' port, and 'xe' SID. The 'Service name' field is empty.

Create User

5. To create a user, run the script file **CreateUser.sql** (shown on the next slide) as the DBA; replace the text **<username>** with the username of the user you want to create and **<password>** with the password you want to assign to the user.
6. Create a connection in **Oracle SQL Developer** for the user created in step 5 to create tables and run queries.

CreateUser.sql Script

```
/* COMP 3311: Script to create a user in Oracle XE */

-- Allow users to be created without the C## prefix
alter session set "_ORACLE_SCRIPT"=true;

-- Create a user and grant required permissions
create user <username> identified by <password> default tablespace users quota unlimited on users;
grant create table, create session, create procedure to <username>;

-- Grant permissions for debugging procedures/functions
grant debug connect session, debug any procedure to <username>;
grant execute on DBMS_DEBUG_JDWP to <username>;

-- PL/SQL procedure to allow a debugging connection to Oracle XE
begin
  dbms_network_acl_admin.append_host_ace
    (host=>'127.0.0.1',
     ace=> sys.xs$ace_type(privilege_list=>sys.XS$NAME_LIST('JDWP'),
      principal_name=>'<username>',
      principal_type=>sys.XS_ACL.PTYPE_DB));
end;
/
```